Band-1 Magic Square of Squares Probe (Collapse 2.29)

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Band-1: 3×3 Magic Square of Distinct Squares — Honest Probe

Honesty stamp. This note does **not** claim a 3×3 magic square of distinct squares was found. It does **not** claim nonexistence is proved by this dash (or by elevating 2025–2026 preprints). It does **not** claim a Clay-millennial solution or a Theory of Everything. **GATE-04** is locked out of scope.

> **Locked OPEN:** existence of a 3×3 magic square whose nine entries are **distinct positive squares** — **OPEN** (community / Wikipedia / Guy D20/D15 lineage).
> **Box:** small-bound exhaustive search under standard center filters — **0 solutions**.
> **2025–2026 nonexistence preprints:** treated as **CLAIMS only** (house rule); **not** elevated to closed.

Abstract

Martin LaBar (1984) asked whether a third-order magic square can have all entries distinct squares; Martin Gardner popularized a cash prize; Richard Guy listed it among unsolved problems in number theory. **No accepted construction is known. No community-accepted impossibility proof is locked here.** Semi-magic / repeated-entry examples exist in the folklore (Parker square context). For $n \geq 4$, magic squares of squares (and more generally of higher powers for large n) are known to exist by modern existence theorems — that **does not** settle $n = 3$.

This Band-1 probe (i) ledgers classical modular / prime-divisor constraints, (ii) verifies several of them on toys on-box, (iii) searches small center bounds for a distinct-squares magic square, and (iv) records **not found**.

Problem (collapse Band-2 item 2.29 → Band-1 probe)

1. Status of the **3×3 magic square of distinct squares**.
2. Cite known **necessary constraints** (not a proof of impossibility).
3. Box search at a **small** honest bound **or** verify a published obstruction on a toy.
4. Keep Clay / ToE / GATE-04 / "solved" off the claim surface.

Literature ledger (web check; not re-proved on box)

Item	Status	Notes
LaBar 1984; Gardner 1996; Guy UPNT	classical open problem	3×3, nine **distinct** squares

Item	Status	Notes
All entries odd; ≡1 mod 8	necessary (lit)	odd squares
Magic sum $S = 3a$ with center a; $S = 72k + 3$ under lit hypotheses	necessary (lit)	Wikipedia / Multimagie summaries
No prime divisor $p = 8k + 3$ of any entry	necessary (lit)	
Center primes all $p = 4k + 1$	necessary (lit)	
Further $p = 8k + 5$ edge/corner linkage rules	necessary (lit)	Wikipedia
Computational searches	no solution in enormous ranges	Multimagie / Chaffin / search notes; centers far beyond this probe
Semi-magic / non-distinct $n \times n$ magic squares of squares, $n \geq 4$	examples exist existence theorems (2024–25 circle-method / constructive lines)	not the open problem **does not close** $n = 3$
2025–2026 “nonexistence” preprints (arXiv/Zenodo claims)	**CLAIM only**	not peer-canon here; do not elevate

Canonical portal. Wikipedia *Magic square of squares*; Christian Boyer, *multimagie.com* (Squares of squares); Guy, *Unsolved Problems in Number Theory*.

Constraint class (collapse surface)

Necessary conditions commonly cited (any genuine solution must obey):

1. **Parity / mod 8:** every entry is an odd square ($\equiv 1 \pmod 8$).
2. **Center:** if the center is a, then every line sums to $S = 3a$; opposite cells through the center sum to $2a$.
3. **Prime form on center:** every prime divisor of the center is $\equiv 1 \pmod 4$.
4. **Forbidden primes on entries:** no prime $p \equiv 3 \pmod 8$ divides any entry.
5. **Edge/corner rules** for primes $p \equiv 5 \pmod 8$ and $p \equiv 3 \pmod 4$ (Wikipedia list).
6. **Scale:** published brute-force / pruned searches push any hypothetical solution to astronomically large entries; this probe does **not** compete with those bounds.

These constrain the **search class**; they are **not** by themselves a completed impossibility proof in this note.

Box experiment

Lab: `experiments/band1/magic_square_squares/`
Code: `probe_mss.py` → `results.json`

Toy constraint checks (pass)

- Odd squares $1..99^2$: all $\equiv 1 \pmod 8$.
- Residues $3, 5 \pmod 8$: never squares on a 500-sample.
- Opposite-pair identity on Lo Shu: pairs through center sum to $2\times$center.
- Center-root filter $p \equiv 1 \pmod 4$ only: applied; $S = 3a \equiv 3 \pmod{72}$ held on the filtered sample.

- Helpers: 25 has no prime $8k+3$; $9=3^2$ does (forbidden class).

Search

For center root c odd with all prime factors $\equiv 1 \pmod 4$, require ≥ 4 representations of $2c^2$ as u^2+v^2 ($u>v>0$), assign four disjoint pairs to the four lines through the center, test all orientations for a full magic square of **nine distinct** positive squares.

Bound	Centers tried (filter)	Centers with ≥ 4 reps	Candidates checked	Solutions
$c \leq 400$	(subset)	—	—	**0**
$c \leq 1200$	146	45	369792	**0**
Brute all odd $c \leq 300$ (no 4k+1 filter)	—	—	—	**0**

Wall ≈ 1.3 s on box. **Not found.**

Verdict

Claim	Result
Found a 3×3 magic square of distinct squares?	**NO**
Proved nonexistence on box?	**NO**
Problem status	**OPEN**
Constraints	Cited + toy-verified (mod 8, center $4k+1$, pair-sum, $8k+3$ forbid class)
Clay / ToE / GATE-04	**out of scope / LOCKED**

References (secondary)

1. Wikipedia, *Magic square of squares* (constraints summary).
2. C. Boyer, Multimagie — Magic squares of squares.
3. R. K. Guy, *Unsolved Problems in Number Theory*.
4. M. LaBar, College Math. J. problem (1984); M. Gardner popularization.
5. Existence for larger n: Rome–Yamagishi / related 2025 *magic squares of powers* literature (context only).

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